# WIRELESS TECHNOLOGY For Medical Applications

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In May 2016 India tabled an ambitious mHealth (Mobile Health) resolution at World Health Organization (WHO), which was supported by over 30 nations. This put India at the forefront of transition towards digital health. Setting up of National Digital Health Authority was another milestone that would help to achieve the stated goal.

The mHealth movement has come as a boon for patients confined to their hospital beds, where their heartbeat is continuously monitored via devices attached to them and displayed with beep sounds. Image this entire burdensome setup converted into a tiny, lightweight temporary body tattoo. The supposed tattoo has an embedded radio transmitter that would monitor and send the patient's vital information to the doctor faster.

This developing field, called mHealth, uses mobile technologies for medical purposes. Use of smartphones for patient care and disease management is a simpler form of the same. Such applications of wireless technologies have led to some very exciting and useful medical innovations.

"Almost all medical devices are becoming connected to the Internet. This is giving users the freedom to utilise data from anywhere. We, as designers, try to ensure every device at least has the option to connect to the Internet," shares Nitesh K. Jangir, co-founder, Coeo Labs.


Wireless technologies in hospitals (Image source: www.summitdata.com)

## Technology on the skin and beyond

Electronics technology is moving towards use inside the body and on the skin. Such technology is known as epidermal electronics. Medical applications for wireless technologies range from diagnostics and monitoring to therapeutics and imaging, and wellness and fitness.

Dr Megha Pruthi, pain management specialist at Venkateshwara Hospital, Delhi, explains health monitoring through wireless devices: "These devices are designed with special wireless sensors. Usually, the technology is patented by makers. The device is embedded in the patient's body during the normal catheterisation procedure. Once embedded, it starts monitoring the patient's vital signs and sending the relevant data to doctors as per the set frequency. In case of any problem, it sends alert messages on phones and emails. This helps to become well prepared to handle the exigency."

Dr Gaurav Chanana, consultant physician, Delhi Pain Management Center, says, "Not just for patient care, wireless technologies are also contributing a great deal in research. Recently, Apple funded a research study run by a group of resident medical students and doctors at Stanford University that enabled real-time dialogue between the doctors and patients using Apple Watch. The same was promoted and encouraged by other doctors as well, and even became a rage on Twitter for a while. No doubt, tech giants like Google, Apple and Microsoft have penetrated deep into the medical system and profoundly transformed the way we take care of our patients."

He adds, "Portable pacemakers have